

Aimee Liu

Cambridge, MA 02139 | aimee003@mit.edu | +1 (929) 409-8283 | [LinkedIn](#) | [Project Portfolio](#)

EDUCATION

Massachusetts Institute of Technology (MIT)

MSc in Mechanical Engineering

Cambridge, MA, USA

Expected: May 2027

Massachusetts Institute of Technology (MIT)

BSc in Mechanical Engineering & Electrical Engineering and Computer Science

Cambridge, MA, USA

May 2025

EXPERIENCE

MIT Biomimetic Robotics Lab

Graduate Researcher

Cambridge, MA

September 2025 – Present

- Develop tactile sensors for a 3-fingered robotic hand, sampling force and contact position at 200 Hz.
- Simulate and train grasping tasks for a robot hand using reinforcement learning in MuJoCo.
- Design wrist mechanism for a robotic humanoid to carry upwards of 18 kg.

ICL Robot Intelligence Lab

Visiting Researcher

London, UK

June 2025 – August 2025

- Simulated locomotion of a bipedal robot (SLIDER) in IsaacGym and Genesis for sim-to-sim validation.
- Trained robot locomotion with reinforcement learning in Python.

PSYONIC Inc.

Mechatronics Intern

San Diego, CA

June 2024 – August 2024

- Programmed robot arm to type on a keyboard and play piano using control algorithms in Python and C++.
- Developed a testing rig for experimenting with finger motors in robotic hand prosthetics (PSYONIC Ability Hand).

MIT Sea Grant

Undergraduate Researcher

Cambridge, MA

February 2024 – May 2024

- Designed and assembled electronic components for an underwater robot (Sea Perch II) and a self-docking robot boat.

MIT Biomechatronics Group

Undergraduate Researcher

Cambridge, MA

June 2022 – May 2023

- Worked on a magnet-based muscle tracking system for robotic prosthetics.
- Reduced code execution time 1,500× and improved calibration accuracy from 1e-6 to 1e-16 radians in Python.

PROJECTS

Autonomous Robot Car

February 2024 – May 2024

- Developed a miniature autonomous car with localization, path planning, and vision algorithms.
- Implemented SLAM techniques, color segmentation, Monte Carlo Localization, directed A* path finding, and pure pursuit.

Bio-Inspired Leaping Robot

September 2023 – December 2023

- Built a custom long-jumping robot with a small team.
- Developed the robot design from a kinematic analysis of the standing long jump.
- Simulated kinematics and dynamics in MATLAB and validated against hardware experiments.

ACTIVITIES & LEADERSHIP

MIT Assistive Technology Club

Vice President / Project Team Lead

May 2023 – May 2025

- Led a team to develop a One-Handed Game Controller and incorporated updates based on feedback from co-designers with disabilities.
- Coordinated training for software and hardware development.

PUBLICATIONS

Nature Sensors – A wireless magnetic implant system for continuous neuromuscular sensing (Co-author)

Accepted

SKILLS

Engineering: CAD (SolidWorks, Fusion360), PCB design (Altium, KiCAD), Embedded Systems (Arduino, STM32, ESP32), 3D Printing, Machining (CNC Lathe, Mill, Water-Jet), Soldering

Machine Learning: PyTorch, TensorFlow, Weights & Biases, Pybullet, MuJoCo, IsaacGym, Genesis, Reinforcement learning

Computer Science: Python, C++, Java, MATLAB, IJulia, ROS/ROS 2, SLAM